Few-Shot Learning Based on Embedded Self-Distillation and Adaptive Wasserstein Distance for Hyperspectral Image Classification

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***Abstract*— Due to the domain shift, it is challenging to achieve ideal experimental results for cross-domain few-shot learning (FSL) in hyperspectral image (HSI) classification. Most existing FSL algorithms are impacted by the limited samples, and they do not effectively leverage the representations from different layers of the network. Therefore, this article proposes an FSL based on embedded self-distillation and adaptive Wasserstein (ESAW-FSL) distance for HSI classification. First, the embedding self-distillation network is proposed in the feature extraction process of the source domain (SD) and the target domain (TD). The embedding self-distillation network utilizes self-distillation from different perspectives to get discriminative features. In the SD, the mask evaluation of embedded features is employed to guarantee the learning of guiding features. Second, a domain adaptation based on adaptive Wasserstein distance is designed to alleviate the domain shift problem between the domains. A lightweight feature correlation network learns the comprehensive cost matrix in the Wasserstein distance adaptively, and the obtained cost matrix helps achieve domain adaptation by an iterative algorithm. Finally, a focal loss based on double softening is adopted in the process of FSL. The probability is double softened to improve the ratio of correctly classifying hard samples. Experiments are conducted on three widely used hyperspectral datasets and compared with six state-of-the-art algorithms. The overall accuracy (OA) and average accuracy (AA) are achieved in multiple experiments, demonstrating the effectiveness of ESAW-FSL.**

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I. Introduction

With the increasing development of artificial intelligence technology and onboard and airborne imaging spectrometers, hyperspectral image (HSI) classification technology has become booming and widespread [1]. In numerous remote sensing image applications [2], [3], the HSI has become a unique branch due to the rich spectral bands [4]. Its increasingly important role in agricultural production [5], land detection [6], geological research [7], ecological monitoring [8], and national defense maintenance [9] has attracted a growing number of researchers' attention [10]. HSI classification technology analyzes the spatial details and spectral bands of an image to learn each pixel's spatial and spectral features [11], thereby determining the class of each pixel. Despite rich spectral information, time-consuming and expensive labels constrain improving classification results [12].

Deep learning methods have recently been widely applied in HSI classification due to their excellent nonlinear representation and feature extraction ability. The convolutional neural network (CNN) can handle the local features effectively. Shang et al. utilized a pyramid coordinate attention mechanism to extract diverse spatial information [13]. However, it is difficult for the CNN to play a role in the non-European domain. The graph convolutional network (GCN) has emerged from people's views due to its excellent performance in the non-European domain. Ding et al. [14] combined the probability graph with the GCN to form a feedback mechanism to alleviate the spatial variability of the HSI. Jia et al. [15] presented a graph-in-graph model and transformed the local and global features into the internal and external information of the superpixels. Constrained by the limited labeled samples, the CNN and GCN are unable to learn good representations of the samples.

The generative adversarial network (GAN) is a data augmentation method that generates high-quality data through adversarial mechanisms. Feng et al. [16] proposed a multicomplementary GAN with contrastive learning to extract intraclass invariant representations. As the networks become deeper, the CNN, GCN, and GAN can lose information during the downsampling process. To obtain long-range dependency

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information, researchers turned to the Vision Transformer (ViT) for global information. Zhao et al. [17] designed a lightweight ViT with a group separable attention mechanism to learn the local and global features. Li et al. [18] used a spectral coordinate Transformer to make features more aggregated. These methods depend highly on labels to achieve satisfactory results. However, obtaining labels is labor-expensive.

With the booming development of artificial intelligence technology, researchers have proposed many methods to avoid the constraints of expensive labels. One of the methods is few-shot learning (FSL). FSL refers to obtaining effective feature learning models through fewer labeled training samples [19]. In the FSL framework, labeled samples are called the support set, generally composed of C classes with K samples in each class. Unlabeled samples are called the query set, and the classification of the query set is mainly based on the similarity with the support set. Therefore, FSL is also known as the C-way K-shot task [20]. Xue et al. [21] proposed a Siamese network based on spectral and spatial HSIs, and the features were constructed by a lightweight network consisting of 1-D convolution and 2-D convolution. Pal et al. [22] introduced a stable prototype network for FSL by combining dropout and DropBlock regularization factors.

However, these methods mentioned above are limited by a few training samples from a single dataset. By introducing other hyperspectral datasets, the number of labeled samples can be increased [23]. However, the direct introduction often brings tribulations to feature representation. The weather circumstances and sensor nonlinearities can lead to changes in spectral reflectance [24]. Even if the data is collected on the same scene, some adjustments are required due to the different acquisition times [25]. To improve the performance of FSL, some researchers have adopted domain adaptation methods to optimize the process [26]. Domain adaptation is a type of transfer learning [27] that can eliminate domain distribution differences between different domains as much as possible. It maps data features to a common feature space, thereby increasing the training data that can be used. The data distribution space of the original dataset is referred to as the target domain (TD), and the data distribution space of the introduced new dataset is referred to as the source domain (SD). Introducing research on the SD data helps better learn the features of the TD data. The feature distribution between the SD and TD is called the domain shift [28]. Eliminating the domain shift as much as possible will make the feature distribution of the SD close to that of the TD, thereby providing more label samples with close distribution for FSL in the TD.

Researchers have proposed multiple ways to eliminate the domain shift based on many approaches. Some try to generate more matching samples through adversarial learning. For example, Shang et al. [29] developed a dual-classifier adversarial paradigm according to local and topological information to enhance the performance of domain-invariant representation. Huang et al. [30] performed concatenation operations on the extracted spatial and spectral features to achieve a better combination. They employed adversarial learning between two classifiers and the generator to align the distribution of the SD and TD [30]. Feng et al. [31] adjusted the distribution by quantifying the correlation between the unaligned TD and the generating data while designing a sampler to prevent network collapse caused by major classes. In comparison, others prefer the domain adaptation theory. For example, Zhang et al. [32] designed intradomain extraction and cross-domain similarity detection blocks to capture the nonlocal relationships. Although the above methods have succeeded greatly, more learning from different perspectives of the feature extraction network can affect the features in both domains.

In response to the above issues, this article proposes an FSL approach based on embedded self-distillation and adaptive Wasserstein distance (ESAW-FSL) for HSI classification. First, the embedding self-distillation network is adopted to extract the intermediate embedding features and hint features of the neural network in the SD and TD. Self-distillation learning of the intermediate embedding features and hint features is performed to enable the transfer of deep feature knowledge and representations from different depths. In self-distillation learning, knowledge transfer of hint features in two domains is carried out from multiple perspectives to enrich the knowledge as much as possible. At the same time, confidence evaluation in the SD is used to learn more instructive feature information. Second, a domain adaptation based on adaptive Wasserstein distance is proposed to alleviate the domain shift. A lightweight feature correlation network is represented to obtain a comprehensive and trainable cost matrix in the Wasserstein distance. Under the guidance of optimal transport theory, the transfer matrix from the SD to the TD is obtained through an iterative algorithm using the obtained cost matrix. The transformation can provide high-quality labeled samples for the TD. Finally, in the process of FSL, a focal loss function based on double softening is designed to evaluate the similarity between the support set and the query set features. The classification information is increased by double softening the probability of the query set. The focal loss function is carried out to assist in classifying hard samples, ultimately improving the classification accuracy of HSIs. The main contributions of the proposed ESAW-FSL are as follows.

1) The embedded self-distillation method is represented in the SD and TD. Knowledge transfer of features is achieved by self-distillation learning of hint features and intermediate embedded features from various perspectives. At the same time, confidence evaluation is performed on the SD to learn more guiding and discriminative features.
2) A domain adaptation based on adaptive Wasserstein distance is adopted. A lightweight feature correlation network obtains a more comprehensive and discriminative cost matrix. With the help of the iterative algorithm, the transport is optimized between hint features in the SD and TD.
3) A focal loss based on double softening is designed to measure the similarity between the support set and the query set. By softening the corresponding probability, the classification information of hard samples in the query set is increased, improving the accuracy of the proposed method.

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The specific arrangement of the remaining work in this article is as follows. Section II provides a detailed introduction to the ESAW-FSL. Section III conducts experiments and analyzes the results of three datasets. Section IV provides a summary.

II. Proposed Method

This article proposes ESAW-FSL to address the issues of limited labeled samples and the domain shift in the cross-domain HSI task. The structure of ESAW-FSL is shown in Fig. 1.

As shown in Fig. 1, the SD and TD data are first passed into two multilayer perceptrons to ensure they share the same spectral band counts. Then, ESAW-FSL puts the SD and TD data into the embedded self-distillation network (ESDN) to learn the intermediate embedding and hint features in their respective domains. Effective knowledge transfer and fusion of features from different layers are achieved through self-distillation learning from multiple perspectives. Subsequently, a domain adaptation based on the adaptive Wasserstein (AWD-DA) distance is adopted to address the domain shift problem between domains. The cost matrix is learned through a lightweight feature correlation network using hint features from two domains. With the help of the Sinkhorn algorithm [33], domain alignment between the SD and TD is gradually achieved through iterations and precious transferred samples are added to relieve the problem of expensive and insufficient labeled samples in the TD. Finally, the hint features are utilized for the FSL. A focal loss function based on double softening (DFL) is proposed to increase the classification information in the query set, thereby improving the accuracy of ESAW-FSL. The core components ESDN, AWD-DA, and DFL will be covered in more detail in Sections II-A– II-C.

A. Embedded Self-Distillation Network

To learn reliable and discriminative features from the labeled samples of the SD and TD, this article proposes an ESDN. In the process of FSL, the SD data is divided into the SD support set S_s and the SD query set Q_s, while the TD data is divided into the TD support set S_t and the TD query set Q_t. Taking SD data as an example, the intermediate embedded features $E(S_s)$ and $E(Q_s)$ are obtained through two feature extraction blocks (FEBs). The embedded feature in the middle consists of two parts

$$\begin{aligned} E(S_s) &= [E_1(S_s), E_2(S_s)] \\ E(Q_s) &= [E_1(Q_s), E_2(Q_s)] \end{aligned} \tag{1}$$

where $E_1(S_s)$ and $E_1(Q_s)$ are obtained by S_s and Q_s, respectively, through the first FEB

$$\begin{aligned} E_1(S_s) &= \mathrm{FEB}_1(S_s) \\ E_1(Q_s) &= \mathrm{FEB}_1(Q_s) \end{aligned} \tag{2}$$

where the FEB consists of a residual block (RB) and a channel attention block (CAB). CAB enhances efficient learning in the channel dimension by using attention mechanisms. $E_2(S_s)$ and $E_2(Q_s)$ are gained by $E_1(S_s)$ and $E_1(Q_s)$, respectively, through the second FEB, respectively,

$$\begin{aligned} E_2(S_s) &= \mathrm{FEB}_2(E_1(S_s)) \\ E_2(Q_s) &= \mathrm{FEB}_2(E_1(Q_s)). \end{aligned} \tag{3}$$

Then, the hint features of the support set $H(S_s)$ and the query set $H(Q_s)$ are acquired through the third FEB

$$\begin{aligned} H(S_s) &= \mathrm{FEB}_3(E_2(S_s)) \\ H(Q_s) &= \mathrm{FEB}_3(E_2(Q_s)). \end{aligned} \tag{4}$$

The similarity matrix M_s is formulated by measuring the similarity between $H(S_s)$ and $H(Q_s)$

$$M_s = F_S(H(S_s), H(Q_s)) \tag{5}$$

where $F_S(\cdot)$ is the similarity function calculated by Euclidean distance. The query set samples can determine the class they belong to by the similarity matrix M_s. Similarly, the TD data features $E_1(S_t)$, $E_1(Q_t)$, $E_2(S_t)$, $E_2(Q_t)$, $H(S_t)$, and $H(Q_t)$ can be attained through the corresponding FEBs. Then, the similarity matrix M_t is acquired by measuring the similarity between $H(S_t)$ and $H(Q_t)$

$$M_t = F_S[H(S_t), H(Q_t)]. \tag{6}$$

The similarity function here also uses the Euclidean distance. After obtaining the similarity matrices M_s and M_t for the two domains, the loss of FSL will be calculated during the forward propagation process of training. The loss function used here will be explained in detail in Section II-C, combining M_s and M_t with the corresponding labels l_s and l_t, respectively.

To fully utilize the intermediate embedding features of ESDN, this article regards the hint features as the teacher model and the intermediate embedded features as the student model for self-distillation learning. Multiple loss functions are proposed from four different perspectives to achieve effective results.

1) Classification Loss: Taking the SD data as an example, the similarity matrix between $E(S_s)$ and $E(Q_s)$ is formulated through the similarity function

$$\begin{aligned} M(E_s) &= [M_1(E_s), M_2(E_s)] \\ &= [F_S(E_1(S_s), E_1(Q_s)), F_S(E_2(S_s), E_2(Q_s))]. \end{aligned} \tag{7}$$

Then, the classification loss $L_C(E_s)$ is given as follows:

$$\begin{aligned} L_C(E_s) &= L_{C1}(E_s) + L_{C2}(E_s) \\ &= L_{\mathrm{CE}}(M_1(E_s), l_s) + L_{\mathrm{CE}}(M_2(E_s), l_s) \end{aligned} \tag{8}$$

where L_{CE} is the cross-entropy loss function and l_s is the label of the query set in the SD.

2) Distillation Loss: The distillation loss between the teacher model and the student model is calculated as follows:

$$\begin{aligned} L_D(E_s) &= L_{D1}(E_s) + L_{D2}(E_s) \\ &= T_s^2 * \left[KL\left(\frac{M_1(E_s)}{T_s}, \frac{M_s}{T_s}\right) + KL\left(\frac{M_2(E_s)}{T_s}, \frac{M_s}{T_s}\right) \right] \end{aligned} \tag{9}$$

where T_s is the SD distillation temperature and KL is the Kullback–Leibler divergence function. The soft knowledge of H_s is imparted to E_s through distillation temperature, thereby improving the learning of E_s features.

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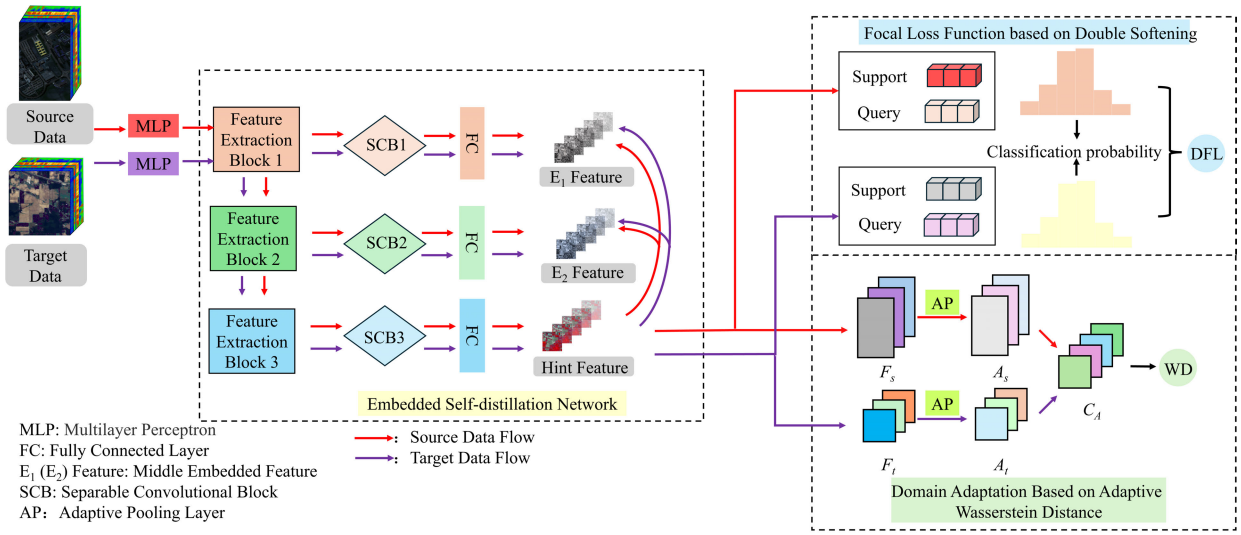


Fig. 1. Overall framework of the ESAW-FSL network.

3) Hint Loss: Affected by FitNet [34] and [35], this article adds a hint loss for the query set to align the distribution of intermediate embedded and hint features. The hint loss is decided by the previous layer results of hint features and intermediate embedded features of the query set. The transform convolution is used to align the results with different dimensions. Let G represent the results of the previous layer in the network, and the corresponding result is represented as

$$\begin{aligned} G_1 &= G(E_1(Q_s)) \\ G_2 &= G(E_2(Q_s)) \\ G_3 &= G(H(Q_s)). \end{aligned} \tag{10}$$

Then, the hint loss can be expressed as

$$\begin{aligned} L_H(E_s) &= L_{H1}(E_s) + L_{H2}(E_s) \\ &= L_2(\mathrm{TC}(G_1), G_3) + L_2(\mathrm{TC}(G_2), G_3) \end{aligned} \tag{11}$$

where TC represents the transformation convolution block and L_2 is the 2-norm function.

4) Confidence Loss: A confidence loss is proposed to get more discriminative SD features. A mask evaluation method is utilized to evaluate the confidence of the SD teacher model. The calculation of the mask is as follows:

$$\mathrm{mask} = \begin{cases} 1, & if\ \max(p(M_s)) > \tau \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $p(M_s)$ represents the classification probability of the similarity matrix M_s. Then, the confidence loss L_{con} is defined as

$$L_{\mathrm{con}} = \mathrm{mask} * \left[(M_s - M_1(E_s))^2 + (M_s - M_2(E_s))^2\right] \tag{13}$$

By evaluating the mask of the similarity matrix M_s, the learning of discriminative features in the SD is enhanced, thereby helping to expand high-quality labeled samples in the TD.

Similarly, in the self-distillation learning process of the TD, there are corresponding loss functions, including the TD classification loss $L_C(E_t)$, the TD distillation loss $L_D(E_t)$, and the TD hint loss $L_H(E_t)$. It is worth noting that there is no confidence loss of the similarity matrix in the TD. Therefore, the loss functions of the embedded self-distillation learning method in the SD and TD are given as follows:

$$\begin{aligned} L^s_{\mathrm{ESD}} &= L_C(E_s) + L_D(E_s) + \lambda * L_H(E_s) + L_{\mathrm{con}} \\ L^t_{\mathrm{ESD}} &= L_C(E_t) + L_D(E_t) + \lambda * L_H(E_t) \\ L_{\mathrm{ESD}} &= L^s_{\mathrm{ESD}} + L^t_{\mathrm{ESD}} \end{aligned} \tag{14}$$

where L^s_{ESD} represents embedded self-distillation learning losses in the SD and L^t_{ESD} represents embedded self-distillation learning losses in the TD. Moreover, λ indicates the weight of the loss function, which is set to 0.005 in this article. L_{ESD} represents the sum of the loss functions in the SD and TD.

B. Domain Adaptation Based on Adaptive Wasserstein Distance

To alleviate the domain shift problem between the SD data and TD data, researchers have proposed Wasserstein distance to guarantee transportation. The cost matrix determined by the data in two domains is essential to the data transportation process. However, determining the cost matrix solely by calculating the cosine distance between data from different domains does not effectively utilize the data. Therefore, a domain adaptation based on adaptive Wasserstein distance is employed to address the domain shift between the SD and the TD data. A lightweight feature correlation network (LFCN) is applied to calculate the cost matrix adaptively, making the transport process more suitable for the distributions of the two-domain data. To be specific, the hint features from each domain in Section II-A are integrated into few-shot features

$$\begin{aligned} F_s &= \mathrm{concat}(H(S_s), H(Q_s)) \\ F_t &= \mathrm{concat}(H(S_t), H(Q_t)) \end{aligned} \tag{15}$$

where concat$(\cdot)$ represents the concatenation operation, which can concatenate two vectors into a more extended vector. F_s and F_t represent the few-shot sample features of the SD

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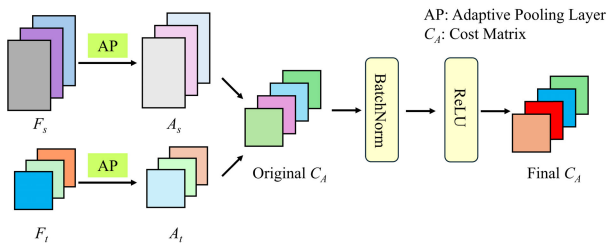


Fig. 2. Framework of the LFCN.

and TD, respectively. F_s and F_t are fed into the LFCN for cost matrix calculation, and the framework of the LFCN is displayed in Fig. 2.

In Fig. 2, AP is the adaptive pooling layer and F_s and F_t are transformed into A_s and A_t through the AP

$$\begin{aligned} A_s &= \text{AdaptivePooling}(F_s) \\ A_t &= \text{AdaptivePooling}(F_t). \end{aligned} \tag{16}$$

Then, A_s and A_t are multiplied to obtain the initial cost matrix, and the final adaptive cost matrix C_A is acquired after batch normalization and an activation layer

$$C_A = \text{act}\left(\text{BN}\left(A_s A_t^T\right)\right) \tag{17}$$

where act $(\cdot)$ is the ReLU activation function. The domain shift between the SD and TD is calculated by employing C_A and Sinkhorn algorithms. According to the optimal transport theory, the Wasserstein distance can be given by the following formula:

$$\begin{aligned} &\min_{T\in\Pi(p,q)} D_W(T) =< T, C_A >_F \\ &s.t. \Pi(p,q) = \left\{T \in \mathbb{R}_+^{n\times n} | T1_n = p, T^T 1_n = q\right\} \end{aligned} \tag{18}$$

where T is the transport matrix containing all the ways of transporting from the SD to the TD, and it is restrained from the SD and TD data. The row sum of T is the marginal distribution p of the SD data, and the column sum is the marginal distribution q of the TD data. $< \cdot, \cdot >_F$ represents the Frobenius dot multiplication operation. The Sinkhorn algorithm adds information entropy about T during the iteration process. First, the initial transition matrix T_0 and the exponential kernel matrix A are calculated

$$T_0 = J_n, A = e^{-\frac{C_A}{\varepsilon}} \tag{19}$$

where n is the size of C_A, J_n is an n-order matrix with all elements equal to 1, and ε is the regularization factor. The row iteration vector U and the column iteration vector V are continuously scaled in the iterations to ensure that T meets the requirements. At the beginning, U and V are initialized as follows:

$$U_0 = J_{1,n}, V_0 = J_{n,1} \tag{20}$$

where n is the size of C_A, $J_{1,n}$ is a row vector with all elements equal to 1, and $J_{n,1}$ is a column vector with all elements equal to 1. In the ith iteration, U and V are updated as follows:

$$U_i = \frac{1}{\text{nAT}_0 V_{i-1}}, V_i = \frac{1}{\text{nAT}_0{}^T U_{i-1}} \tag{21}$$

where U_{i-1} and V_{i-1} are the output in the previous iteration. After the iteration, the final transport matrix T is calculated as follows:

$$T = \text{diag}(U_{\text{fin}}) T_0 \text{diag}(V_{\text{fin}}) \tag{22}$$

where U_{fin} and V_{fin} are the results after reaching the maximum iterations or satisfying the stopping condition. And diag(U_{fin}) and diag(V_{fin}) are diagonal matrices constructed from U_{fin} and V_{fin}. Then, the adaptive Wasserstein distance D_W can be obtained based on (21). Domain adaptation can be achieved from the SD to the TD by continuously adjusting T to minimize the Wasserstein distance.

The calculation process of the adaptive Wasserstein distance based on the Sinkhorn algorithm is given by Algorithm 1.

Algorithm 1 Adaptive Wasserstein Distance

Input: Adaptive cost matrix C_A, size of C_A n, number of iterations N, stopping threshold δ.
Output: Adaptive Wasserstein distance D_W.

```
Calculate the initial transport matrix T_0 and the exponential kernel matrix A based on Eq. (19);
Initialize the row iteration vector U and the column iteration vector V based on Eq. (20);
for i = 1, 2, ..., N do
    Update U_i and V_i based on Eq. (21);
    if |U_i − U_{i−1}| < δ then
        U_fin = U_i;
        V_fin = V_i;
    end if
end for
Calculate the final transport matrix T based on Eq. (22);
Get the adaptive Wasserstein distance D_W based on Eq. (18).
return D_W.
```

C. Focal Loss Function Based on Double Softening

In Section II-A, it is mentioned that during the forward propagation process of training, the loss of FSL is calculated by M_s and M_t with corresponding labels in the SD and TD. In the classification process, the class of the samples is determined based on the first classification information. However, some samples' true class corresponds to the second classification information, called hard samples. This situation affects the performance of the network. Therefore, a focal loss based on double softening (DFL) is designed, which softens the corresponding classification probability to increase classification information for hard samples of the query set. Thus, DFL can relieve the misclassification situation of hard samples and enhance the accuracy of the network. Taking the SD data as an example, the softening probability of M_s is formulated through SoftMax as follows:

$$sp_s = \text{SoftMax}(\alpha \text{SoftMax}(M_s)) \tag{23}$$

where $\alpha > 1$ is the softening factor, which can help modulate the softening situation. Compared with the results calculated

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TABLE I
NETWORK STRUCTURE AND PARAMETERS OF THE FEB

Module	Layer	Kernel Size	Stride	Padding	BN
RB	Conv 1	3×3×3	1×1×1	1×1×1	Yes
	Conv 2	3×3×3	1×1×1	1×1×1	Yes
	Conv 3	3×3×3	1×1×1	1×1×1	Yes
	Act 1	Activate the sum of the results of Conv 1 and Conv 3 by ReLU.			
CAB	AP	-			
	FC 1	Reduce the number of channels by half.			
	Act 2	Activate the results of FC 1 by ReLU.			
	FC 2	Double the number of channels.			
	Act 3	Activate the results of FC 2 by Sigmoid.			
	Mul	Multiply the result of Act 3 with the CAB input.			

BN is the batch normalization, Conv is the convolutional layer, Act is the activation layer, AP is the adaptive pooling layer, FC is the fully connected layer, and Mul is the multiplication operation.

TABLE II
NETWORK STRUCTURE AND PARAMETERS OF THE SCB

Layer	Kernel Size	Stride	Padding	BN
Conv 1	3×3×3	1×1×1	1×1×1	No
Conv 2	1×1×1	-	-	Yes
Act 1	Activate the results of Conv 2 by ReLU.			
Conv 3	3×3×3	1×1×1	1×1×1	No
Conv 4	1×1×1	-	-	Yes
Act 2	Activate the results of Conv 4 by ReLU.			
Output	Add the input and the result of Act 2 to get the output.			

BN is the batch normalization, Conv is the convolutional layer, and Act is the activation layer.

TABLE III
NETWORK STRUCTURE AND PARAMETERS OF THE TC

Object	Layer	Kernel Size	Stride	Padding	BN
SD data	Conv 1	1×1×1	-	-	Yes
	Act 1	Activate the results of Conv 1 by ReLU.			
TD data	Conv 2	1×1×1	-	-	Yes
	Act 2	Activate the results of Conv 2 by ReLU.			
	Mul 1	Multiply the result of Act 1 with that of Act 2.			
	Mul 2	Multiply the result of Mul 1 with that of Act 1.			
	FC	-			

BN is the batch normalization, Conv is the convolutional layer, Act is the activation layer, Mul is the multiplication operation, and FC is the fully connected layer.

only once by SoftMax, sp_s contains more secondary classification information. Thus, sp_s can increase the probability of hard samples being classified as secondary classes. Then, the focal loss of SD data using sp_s is learned by

$$\mathrm{DFL}(sp_s) = -\beta(1 - sp_s)^{\gamma} \log(sp_s) \tag{24}$$

where β is the balance factor used to handle cases of imbalanced samples and γ is the regulatory factor controlling the learning performance of hard samples. The DFL of the TD data has the same form as $\mathrm{DFL}(\mathrm{sp}_s)$, which is represented as follows:

$$\mathrm{DFL}(sp_t) = -\beta(1 - sp_t)^{\gamma} \log(sp_t) \tag{25}$$

where sp_t is the softening probability of M_t. Then, the DFL of the two domains is represented as follows:

$$\mathrm{DFL} = \mathrm{DFL}(sp_s) + \mathrm{DFL}(sp_t). \tag{26}$$

By improving the secondary classification information, DFL can assist in learning information in hint features and enhance the results of the network.

In summary, the loss function during network training is represented as follows:

$$L = \mathrm{DFL} + \omega_1 L_{\mathrm{ESD}} + \omega_2 D_W(T) \tag{27}$$

where ω_1 and ω_2 are the weights of L_{ESD} and $D_W(T)$, respectively.

D. Network Parameters

The proposed method utilizes multiple network blocks for feature learning and representation. The specific parameters of the network block are introduced as follows.

1) FEB: In Section II-A, the FEB consists of RB and CAB, and the specific structure and parameters are shown in Table I

2) Separable Convolutional Blocks: In Fig. 1, ESAW-FSL learns features in more detail through separable convolutional blocks (SCBs). The specific structure and parameters of SCBs are shown in Table II.

3) Transformation Convolution Block TC: In Section II-A, the hint loss is calculated by aligning the previous layer results of features with different dimensions through TC. The specific structure and parameters of the TC are shown in Table III.

The basic processes of ESAW-FSL are shown in Algorithm 2.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Description

The experiments are conducted on four datasets widely used in cross-domain HSI classification tasks to verify the effectiveness of the proposed method in this article. These four datasets are the Chikusei (CH), the Indian Pines (IP), the Salinas (SA), and the Pavia University (PU). The CH dataset is selected as the SD dataset, and the others are selected as the TD dataset. The detailed introduction of the four datasets is as follows.

1) CH Dataset: This dataset was captured by the Headwall Hyperspec-VNIR-C imaging sensor in the agricultural and urban areas of Chikusei, Ibaraki, Japan. The imaging sensor selects 128 bands in the spectral range of 363–1018 nm at a height of 5 m from the ground. The high-resolution color image and hyperspectral data consisted of 2517 × 2335 pixels, and 19 classes of ground conditions were collected through field investigation and visual inspection. Table IV shows the related colors, the land-cover information, and the sample number for specific classes for the CH dataset. Fig. 3 shows the false-color image and the ground-truth map of the CH dataset.

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Algorithm 2 ESAW-FSL

Input: The SD data X_s, the TD data X_t, the maximum number of iterations I.

Output: Classification map Y.

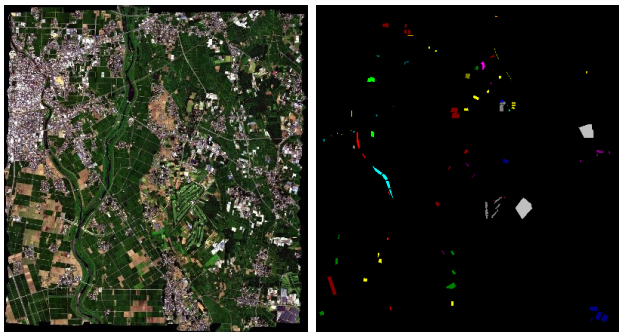
```
Divide the X_t into training set Tr and testing set Te;
Construct support and query set data from X_s and X_t;
for i = 1, 2, ..., I do
    Obtain E(S_t), E(Q_t), H(S_t), H(Q_t), E(S_t), E(Q_t), H(S_t), and H(Q_t) through ESDN;
    if the features are from the SD then
        Calculate the L^s_ESD according to Eq. (8), (9), (11), and (13);
    else
        Calculate the L^t_ESD;
    end if
    Use LFCN to calculate the adaptive cost matrix C_A;
    Calculate the D_W(T) based on Algorithm 1;
    Calculate DFL according to Eq. (26);
    Calculate the total loss according to Eq. (27) and update the network;
end for
Extract features from the Tr and Te using the trained network;
Obtain the classification map Y of the Te through clustering.
return Y.
```

TABLE IV

COLOR, LAND COVER, AND NUMBER OF SAMPLES FOR THE CH DATASET

Class	Color	Land cover class	Samples
0		Background	5799603
1		Water	2345
2		Bare soil (school)	2859
3		Bare soil (park)	286
4		Bare soil (farmland)	4852
5		Natural plants	4297
6		Weeds in farmland	1108
7		Forest	20516
8		Grass	6515
9		Rice field (groim)	13369
10		Rice field (first stage)	1268
11		Row crops	5961
12		Plastic house	2193
13		Manmade (nan-dark)	1220
14		Manmade (dark)	7664
15		Manmade (blue)	431
16		Manmade (red)	222
17		Manmade grass	4040
18		Asphalt	801
19		Paved ground	145
		Total	5877195

2) IP Dataset: This dataset is an image of the Purdue University Agricultural Farm and its surrounding areas northwest of West Lafayette, United States, obtained by the airborne visible/infrared imaging spectrometer (AVIRIS). The dataset comprises 145 × 145 pixels and 220 spectral reflection bands with a wavelength range of 0.4–2.5 microns. This area includes



(a) (b)

Fig. 3. CH dataset. (a) False-color image. (b) Ground-truth map.

TABLE V

COLOR, LAND COVER, AND NUMBER OF SAMPLES FOR THE IP DATASET

Class	Color	Land cover class	Samples
0		Background	10776
1		Alfalfa	46
2		Corn notill	1428
3		Corn min	830
4		Corn	237
5		Grass/Pasture	483
6		Grass/Trees	730
7		Grass/pasture-mowed	28
8		Hay windrowed	478
9		Oats	20
10		Soybeans notill	972
11		Soybeans min	2455
12		Soybeans clean	593
13		Wheat	205
14		Woods	1265
15		Bldg Grass Tree Drives	386
16		Stone steel towers	93
		Total	21025

the main parts of the Indian Creek and Pine Creek watersheds. After removing the 104–108, 150–163, and 220th bands affected by water surface absorption and ozone interference, the final dataset contains 200 bands. This test dataset contains 16 classes with 10 249 labeled pixels, including grasslands, forests, farmland, and other vegetation. Table V shows the related colors, the land-cover information, and the sample number for specific classes for the IP dataset.

3) PU Dataset: This dataset was captured by the reflective optics system imaging spectrometer (ROSIS) over the campus of the University of Pavia in Italy. The dataset image comprises 610 × 340 pixels, with 115 spectral reflection bands selected in the wavelength range of 0.43–0.86 microns. After removing 12 noisy bands, the final dataset contains 103 spectral reflection bands. This test dataset contains nine classes of land objects and 42 776 labeled pixels. Table VI shows the corresponding colors, land-cover information, and sample size of specific samples for the PU dataset.

4) SA Dataset: This dataset was collected by AVIRIS sensors over Salinas Valley, California, covering an area of 512 × 217 pixels. Like the scene of the IP dataset, the number of spectral bands in this dataset decreased from 224 to 220 after removing the spectral reflection bands absorbed by the water

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TABLE VI
COLOR, LAND COVER, AND NUMBER OF SAMPLES FOR THE PU DATASET

Class	Color	Land cover class	Samples
0		Background	164624
1		Asphalt	6631
2		Meadows	18649
3		Gravel	2099
4		Trees	3064
5		Painted metal sheets	1345
6		Bare Soil	5029
7		Bitumen	1330
8		Self-Blocking Bricks	3682
9		Shadows	947
		Total	207400

TABLE VII
COLOR, LAND COVER, AND NUMBER OF SAMPLES FOR THE SA DATASET

Class	Color	Land cover class	Samples
0		Background	10895
1		Broccoli green weeds 1	2009
2		Broccoli green weeds 2	3726
3		Fallow	1976
4		Fallow rough plow	1394
5		Fallow smooth	2678
6		Stubble	3959
7		Celery	3579
8		Grapes untrained	11271
9		Soil vineyard develop	6203
10		Corn senesced green weeds	3278
11		Lettuce romaine 4wk	1068
12		Lettuce romaine 5wk	1927
13		Lettuce romaine 6wk	916
14		Lettuce romaine 7wk	1070
15		Vinyard untrained	7268
16		Vinyard vertical trellis	1807
		Total	65024

surface. This dataset includes 16 classes: vegetables, exposed soil, and vineyards. Table VII shows the related colors, land-cover information, and the sample number for specific classes for the SA dataset.

B. Evaluation Metrics

This experiment employs three evaluation metrics—the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (KC)—to assess the classification results. OA highlights more correctly classified samples by referring to the ratio of correctly classified samples with all samples during the testing phase. AA emphasizes the classification accuracy of each class by referring to the average ratio of correctly classified samples in each class. KC concentrates on unbalanced data and is computed by the confusion matrix. It can effectively evaluate various methods on datasets with imbalanced samples.

C. Comparison Algorithms and Experimental Settings

This article compares ESAW-FSL with 6 state-of-the-art algorithms to verify the effectiveness of the proposed method, including the 3-D spatial residual network (SSRN) [36], deep FSL (DFSL) [37], deep cross-domain FSL (DCFSL) [38], FSL with class covariance metric (CMFSL) [39], and multistage relationship network with dual metric (DM-MRN) [40] and Gia-CFSL [32]. Except for SSRN and DM-MRN, the remaining algorithms are cross-domain FSL methods for HSI classification. Therefore, only SSRN and DM-MRN are directly tested on the TD datasets, while the remaining methods are trained using the CH dataset as the SD dataset and then tested on three TD datasets. As set in DCFSL and Gia-CFSL, classes with a sample size of over 200 are selected from the CH dataset to build the source data, and both the SD and TD data are processed into 9×9 patches for experiments. As for the training process on the SD and TD datasets, the spectral band counts of the SD and TD datasets are different. To make the spectral band counts of two domains the same, this article adjusts the spectral band counts of the data from both fields through two multilayer perceptrons. After using the multilayer perceptrons, the spectral band counts of both domain datasets are the same, which is 128 in this article. The categories of the SD and TD datasets are different, and the model may output different categories without treatment. Instead of using a shared output header, this article reduces the number of categories in the SD data during the data-processing stage. The number of categories in the TD dataset determines that in the SD dataset. Specifically, when the TD datasets are the IP and SA datasets with 16 categories, the SD dataset selects data from 16 categories in the experiment. When the TD dataset is the PU dataset with nine categories, the SD dataset selects data from nine categories. In the FSL task of the SD and TD, the number of samples in the support set is set to 1, and the number of samples in the query set is set to 19. The number of training iterations in the experiment is 10 000 and the learning rate is 0.0005. Xavier normalization is used to initialize convolutional kernels, while Kaiming normalization is used to initialize linear layers. All experiments are conducted under the PyTorch framework deployed on CPU Xeon E5-2630 and GPU NVIDIA RTX2080Ti. Moreover, the experiments are repeated ten times to prevent accidental phenomena.

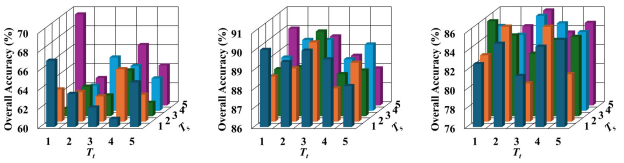
To demonstrate the impact of different numbers of FEBs, the experiment is conducted with 2, 3, and 4 FEBs. The classification and the efficiency are the metrics to decide the count of the FEBs that are used in the article. The efficiency is shown by the training time of a single iteration and the test time. It is noticed that ESAW-FSL does not distill the information located in the second block to the first one. A related experiment is also conducted to explain the reason. The experimental results are shown in Table VIII.

In Table VIII, the experiment that distills the information in the second block to the first one is represented as "2->1." The classification results are improved with more FEBs, which shows that the FEB can extract information effectively. However, when the number of FEBs is 4, the training time is improved by 0.5, 0.65, and 0.39 s compared with the results with 3 FEBs. In other words, time consumption has increased by at least 35.78 %. At the same time, the OA results are improved by 0.39 %, 0.21 %, and 0.47%. The OA results of four blocks do not show a significant improvement compared to three blocks, but the time cost of four blocks is more.

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TABLE VIII
RESULTS OF DIFFERENT NUMBER OF FEBS AND DIFFERENT DISTILLATION APPROACHES (%)

Dataset	Metric	2 FEBs	3 FEBs	4 FEBs	2->1
IP	OA	67.87±4.38	69.01±4.11	**69.34±4.60**	68.69±4.28
	AA	78.41±2.31	80.43±1.62	**80.94±3.63**	79.69±3.10
	KA	63.37±4.57	65.23±4.38	**65.87±5.17**	64.15±4.73
	Training(s)	**0.82**	1.11	1.61	1.27
	Test(s)	**3.18**	4.70	5.83	5.02
SA	OA	90.27±1.37	90.73±1.31	**91.16±1.42**	90.28±1.44
	AA	94.60±0.88	94.91±0.85	**95.25±0.87**	94.79±0.92
	KA	89.31±1.49	89.71±1.44	**90.12±1.51**	89.20±1.58
	Training(s)	**1.14**	1.41	2.06	1.80
	Test(s)	**17.20**	22.08	24.53	25.68
PU	OA	83.89±2.32	85.70±2.44	**86.17±2.56**	85.59±2.61
	AA	80.73±2.42	85.79±2.17	**86.13±2.25**	85.54±2.20
	KA	80.99±3.10	81.15±3.07	**81.70±3.25**	80.42±3.16
	Training(s)	**0.80**	1.09	1.48	1.25
	Test(s)	**12.07**	17.45	17.61	17.55



(a) (b) (c)

Fig. 4. Experimental results of different T_s and T_t on three datasets. (a) IP. (b) SA. (c) PU.

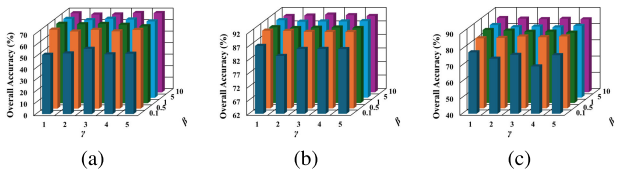
Therefore, the number of feature extraction is determined to be 3. After adding the distillation information from the second block to the first block, the OA results are decreased by 1.26%, 0.67%, and 0.11%, and time consumption increases. Therefore, this article does not distill the information from the second block to the first block.

Compared with the experimental results of two blocks, the results of three blocks have shown improvement, while the efficiency of three blocks has not decreased much. However, compared with the experimental results of three blocks, the experimental results of four blocks are not improved obviously, but time consumption increases remarkably. On the other hand, as the number of blocks increases, the distillation method also increases, which may decrease the results.

In ESDN, SD and TD features are integrated through embedded self-distillation learning. To explore the best performance in two domains, the experiments are set up to select the optimal SD distillation temperature T_s and TD distillation temperature T_t in (13). Both distillation temperatures are selected from [1, 2, 3, 4, 5]. The experimental results on three datasets are shown in Fig. 4.

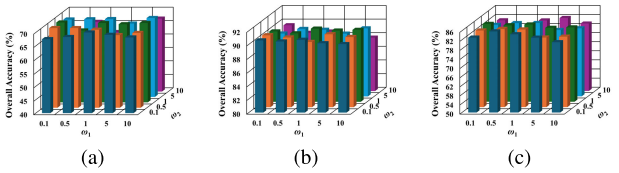
According to the experimental results in Fig. 4, the optimal T_s and T_t on the IP, SA, and PU datasets are $\{T_s = 5, T_t = 1\}$, $\{T_s = 3, T_t = 3\}$, and $\{T_s = 4, T_t = 2\}$, respectively.

In the process of calculating DFL, β and γ affect the learning situation for hard samples. To explore the optimal values of β and γ, the experiments are set up to select the optimal β from [0.1, 0.5, 1, 5, 10] and γ from [1, 2, 3, 4, 5], respectively. The experimental results on three datasets are shown in Fig. 5.



(a) (b) (c)

Fig. 5. Experimental results of different β and γ on three datasets. (a) IP. (b) SA. (c) PU.



(a) (b) (c)

Fig. 6. Experimental results of different ω_1 and ω_2 on three datasets. (a) IP. (b) SA. (c) PU.

TABLE IX
RESULTS OF ABLATION EXPERIMENTS ON THE THREE DATASETS (%)

Dataset	Metric	ESAW-FSL	w/o ESDN	w/o AWD-DA	w/o DFL
IP	OA	**70.15±2.30**	52.70±2.29	65.09±2.13	64.79±1.82
	AA	**80.63±1.73**	66.72±2.23	80.13±1.82	80.19±1.33
	KC	**66.73±2.58**	47.11±2.80	61.28±2.22	61.04±1.95
SA	OA	**90.95±0.38**	85.48±0.99	89.34±0.62	90.05±0.85
	AA	**95.61±0.37**	92.19±0.54	94.02±0.54	94.52±0.34
	KC	**90.48±0.43**	83.88±1.09	88.15±0.69	88.94±0.94
PU	OA	**85.70±2.44**	77.47±2.51	85.44±2.05	84.65±2.05
	AA	**85.79±2.17**	79.64±1.30	85.19±1.75	85.83±1.85
	KC	**81.15±3.07**	70.89±2.92	81.07±2.55	80.21±2.48

According to the experimental results in Fig. 5, the optimal β and γ on the IP, SA, and PU datasets are $\{\beta = 1, \gamma = 1\}$, $\{\beta = 0.5, \gamma = 1\}$, and $\{\beta = 10, \gamma = 1\}$, respectively.

To explore the optimal values of ω_1 and ω_2 in (27), the experiments are set up to select the optimal ω_1 and ω_2 from [0.1, 0.5, 1, 5, 10]. The experimental results with different ω_1 and ω_2 are shown in Fig. 6.

According to the experimental results in Fig. 6, the optimal ω_1 and ω_2 on the IP, SA, and PU datasets are $\{\omega_1 = 1, \omega_2 = 1\}$, $\{\omega_1 = 1, \omega_2 = 1\}$, and $\{\omega_1 = 0.5, \omega_2 = 0.1\}$, respectively.

D. Ablation Study

A series of experiments are designed on three datasets to demonstrate the effectiveness of each component in ESAW-FSL. Specifically, removing the ESDN method means that the SD and TD data are directly learned with FSL through FEBs and SCBs. When removing the DFL, the cross-entropy loss is considered. The results of the ablation experiments are shown in Table IX.

Table IX shows that after removing each component of ESAW-FSL, the OA, AA, and KC on the three datasets all decreased, indicating the effectiveness of ESDN, AWD-DA, and DFL. When the ESDN is removed, the OA results are decreased by 17.45%, 5.47%, and 8.23%, the AA results are decreased by 13.91%, 3.472%, and 6.15%, and the KC results are decreased by 19.62%, 6.60%, and 10.26% on the

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TABLE X
CLASSIFICATION RESULTS ON THE IP DATASET (%, 5 LABELED SAMPLES PER CLASS)

Class	SSRN	DFSL+NN	DCFSL	CMFSL	DM-MRN	Gia-FSL	ESAW-FSL
1	52.55±23.91	95.87±2.99	97.32±4.56	99.02±2.24	99.02±1.62	93.90±5.26	**99.27±1.12**
2	**67.96±12.57**	40.77±10.78	45.09±9.32	50.42±11.68	48.23±11.43	50.13±9.01	60.13±7.72
3	57.28±20.90	**69.63±9.32**	55.49±9.29	52.67±5.61	48.96±9.79	51.99±15.03	67.18±4.40
4	62.86±17.92	77.85±4.81	78.79±12.32	**90.22±8.10**	84.53±10.93	77.84±13.71	79.03±3.70
5	83.09±15.08	80.37±8.59	74.21±8.03	67.97±17.34	72.13±13.64	71.80±8.69	**86.69±6.56**
6	86.52±12.43	93.70±3.80	89.88±6.42	**95.67±1.83**	95.49±2.08	83.42±7.29	68.61±11.03
7	54.19±21.54	**100.00±0.00**	99.13±1.74	99.57±1.30	**100.00±0.00**	96.96±6.16	99.57±1.30
8	95.93±6.80	88.70±6.76	86.51±9.62	97.72±1.84	**98.10±2.35**	88.88±8.80	80.66±12.00
9	21.39±18.67	**100.00±0.00**	99.33±2.00	**100.0±0.00**	**100.00±0.00**	**100.0±0.00**	99.33±2.00
10	**71.68±11.56**	67.18±8.44	59.21±11.34	70.81±4.43	65.56±10.63	62.82±4.55	60.35±6.17
11	66.99±11.86	52.97±8.63	57.95±12.85	50.45±13.59	56.22±14.14	66.20±6.52	**67.86±4.23**
12	60.49±15.06	**61.65±10.98**	47.69±11.96	52.14±7.16	47.13±7.31	43.15±11.16	53.23±5.39
13	80.99±10.83	94.78±5.80	99.45±0.69	98.6±1.96	95.00±6.15	99.20±1.00	**99.85±0.23**
14	**94.19±3.28**	88.15±4.41	88.63±5.65	91.54±3.93	91.20±5.11	85.60±9.69	85.69±3.22
15	53.70±15.43	78.52±7.63	71.97±5.46	73.60±8.00	77.19±11.64	65.43±12.03	**79.66±7.32**
16	72.91±13.02	97.53±3.3	**99.09±1.42**	98.18±3.45	98.30±2.50	97.73±1.97	98.98±1.29
OA	66.78±3.95	67.86±1.41	66.13±3.89	67.51±3.41	67.60±3.24	67.48±3.13	**69.95±2.30**
AA	67.67±4.53	79.48±1.45	78.11±1.72	80.14±1.13	79.82±2.31	77.19±2.44	**80.63±1.73**
KC	62.49±4.12	64.09±1.54	61.92±4.18	63.81±3.57	63.73±3.45	63.21±3.46	**66.73±2.58**

TABLE XI
CLASSIFICATION RESULTS ON THE SA DATASET (%, 5 LABELED SAMPLES PER CLASS)

Class	SSRN	DFSL+NN	DCFSL	CMFSL	DM-MRN	Gia-FSL	ESAW-FSL
1	80.95±5.81	**99.68±0.19**	99.22±0.74	99.54±0.95	99.56±0.96	98.46±1.36	99.37±1.06
2	99.52±0.88	99.77±0.32	98.93±2.41	**99.88±0.04**	99.70±0.88	98.54±3.29	99.76±0.27
3	90.40±13.12	96.41±6.27	92.46±9.22	99.22±1.62	97.39±3.63	89.64±10.61	**99.45±0.79**
4	83.62±10.12	98.84±0.79	99.67±0.21	97.50±4.17	**99.90±0.21**	99.31±0.76	99.40±0.43
5	**96.60±3.95**	94.16±4.07	92.49±2.51	93.09±5.86	92.55±4.23	89.27±5.03	95.14±1.26
6	99.07±1.03	99.28±0.56	98.69±1.37	98.87±1.37	98.94±2.08	98.47±1.61	**99.99±0.02**
7	**99.44±0.76**	99.07±0.45	99.25±1.29	99.37±0.99	97.35±4.68	98.36±3.34	98.98±1.14
8	82.70±7.61	66.60±9.32	74.26±11.98	68.80±12.40	69.86±6.70	76.33±9.75	**83.62±2.54**
9	98.77±0.83	99.29±0.31	99.19±1.54	99.37±0.26	99.51±1.30	98.44±1.92	**99.97±0.04**
10	**94.72±4.10**	85.38±7.50	85.50±5.22	92.13±4.22	94.62±1.88	78.20±13.97	89.16±2.46
11	94.61±5.03	98.50±0.66	98.48±1.24	99.09±1.02	98.19±3.68	96.00±2.37	**99.24±0.85**
12	95.97±4.51	**99.70±0.57**	99.59±0.70	97.87±1.66	98.26±2.08	98.44±2.78	99.31±0.60
13	87.15±14.20	**99.98±0.07**	98.89±0.75	96.90±4.83	97.27±2.69	98.46±1.02	98.90±0.99
14	87.92±10.47	98.52±0.91	98.69±1.31	96.37±2.87	96.64±3.85	98.08±2.32	98.04±1.16
15	70.97±13.45	77.30±9.46	72.95±7.26	**81.84±7.99**	79.48±7.07	72.60±6.54	70.55±3.96
16	85.90±9.46	98.17±0.79	91.89±4.82	98.06±2.36	98.81±2.40	89.96±6.21	**98.89±064**
OA	86.90±3.25	88.31±1.24	88.77±2.22	89.63±1.96	89.57±1.63	89.52±2.11	**90.95±0.38**
AA	90.52±1.91	94.42±0.59	93.76±0.95	94.89±0.91	94.88±0.90	92.46±1.22	**95.61±0.37**
KC	85.46±3.53	87.03±1.36	87.52±2.43	88.51±2.15	88.44±1.80	88.33±2.31	**90.48±0.43**

IP, SA, and PU datasets, respectively. Further observation of the experimental results shows that when the ESDN network is removed, the results on the three datasets decrease the most. This is because the ESDN network efficiently combines feature information from different layers through embedded self-distillation learning.

E. Classification Results and Analysis

The results of 6 state-of-the-art algorithms and ESAW-FSL on the IP dataset as the TD dataset are shown in Table X. In this part, the highest accuracy of each class is highlighted in bold, while the suboptimal accuracy is underlined.

As shown in Table X, when only five samples are used for training in each class, ESAW-FSL achieved the best results on the OA, AA, and KC metrics, which are improved by 2.09%, 0.81%, and 2.64% compared with the second-best, respectively. Compared with FSL algorithms such as SSRN and DM-MRN, the ESAW-FSL helps TD samples establish clearer boundaries between classes by introducing SD information. Compared with cross-domain FSL algorithms, ESAW-FSL better characterizes the features of the SD and TD from different perspectives through embedded self-distillation learning.

The experimental results of six state-of-the-art algorithms and ESAW-FSL on the SA dataset as the TD dataset are shown in Table XI.

As shown in Table XI, when only 5 samples are used for training on the SA dataset for each class, ESAW-FSL achieved the best results on the OA, AA, and KC, which are improved

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TABLE XII
CLASSIFICATION RESULTS ON THE PU DATASET (%, 5 LABELED SAMPLES PER CLASS)

Class	SSRN	DFSL+NN	DCFSL	CMFSL	DM-MRN	Gia-FSL	ESAW-FSL
1	60.54±16.58	64.98±6.14	76.75±14.83	75.35±4.45	74.45±8.17	79.56±11.54	**80.38±6.27**
2	90.31±5.90	80.45±5.16	87.56±8.66	78.07±7.43	69.22±15.46	90.16±3.75	**92.05±4.67**
3	49.98±18.40	**81.74±7.42**	63.77±12.26	76.22±8.81	80.48±11.03	69.78±9.43	74.94±11.5
4	82.88±18.66	91.80±2.05	92.38±4.17	83.18±5.99	93.26±2.37	89.01±3.75	86.17±6.28
5	98.45±1.37	99.93±0.09	99.01±1.09	83.97±7.76	**99.95±0.07**	99.47±1.37	99.67±0.45
6	81.91±13.85	83.39±8.05	72.44±9.01	**99.76±0.46**	83.44±10.22	71.62±9.96	71.53±7.76
7	52.73±24.20	91.27±3.52	79.71±7.04	88.13±6.76	**97.52±2.36**	87.38±4.45	92.55±4.32
8	52.07±15.81	68.10±10.12	65.45±10.61	**99.15±0.65**	75.93±11.08	76.15±13.12	77.68±11.56
9	67.70±19.06	89.48±3.91	98.16±1.98	70.11±11.57	**99.21±1.09**	97.49±3.32	97.18±2.14
OA	69.09±7.86	79.35±1.54	81.73±4.27	82.96±3.79	77.06±6.30	84.42±3.09	**85.70±2.44**
AA	71.84±3.95	83.46±1.13	81.69±2.19	85.00±1.27	85.02±2.29	84.52±2.32	**85.79±2.17**
KC	60.78±9.02	73.66±1.81	76.26±5.02	78.35±4.45	71.49±7.05	79.54±3.88	**81.15±3.07**

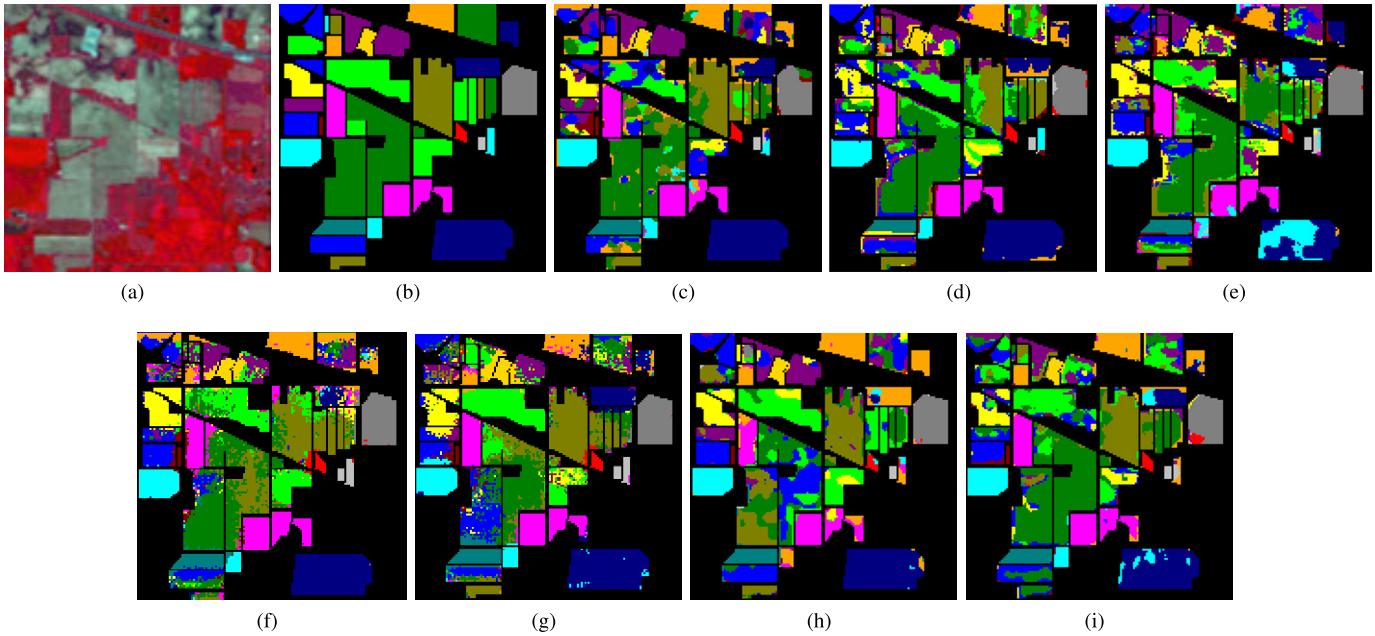


Fig. 7. Visualization maps of various algorithms on the IP dataset. (a) False-color image. (b) Ground truth. (c) SSRN. (d) DFSL+NN. (e) DCFSL. (f) CMFSL. (g) DM-MRN. (h) Gia-FSL. (i) ESAW-FSL.

by 1.32%, 0.72%, and 1.972%, respectively. This indicates that the ESAW-FSL algorithm uses AWD-DA to transform SD features into the TD, contributing to the feature learning and classification of TD samples. The ESAW-FSL algorithm achieves the highest classification accuracy in 6 classes. Compared with other algorithms, the ESAW-FSL algorithm learns the feature information of the ESDN at different layers from numerous perspectives, improving the learning of network information.

The results of 6 state-of-the-art algorithms and ESAW-FSL on the PU dataset are shown in Table XII.

As shown in Table XII, when only 5 samples are used for training for each class on the PU dataset, ESAW-FSL achieves the highest accuracies in OA, AA, and KC, which are 1.28%, 0.77%, and 1.61% higher than the second-best, respectively. This is attributed to the fact that the ESAW-FSL algorithm adopts the DFL loss function during the FSL process in both domains, which softens the classification probability of samples by double softening, increases the classification information, and improves the classification process of hard samples. From the details of each category, the ESAW-FSL algorithm achieves the highest classification accuracy in the Asphalt and Meadows classes and achieves the suboptimal classification accuracy in the Bitumen and Self-Blocking Bricks classes.

The visualization results of 6 state-of-the-art algorithms and ESAW-FSL on the IP dataset as the TD dataset are shown in Fig. 7.

As shown in Fig. 7(b), labeled samples are displayed using pixels of different colors, while unlabeled samples are represented as backgrounds using black pixels. The class information corresponding to these colors can be queried in Table V. In Fig. 7(c), the classification results of SSRN are easily influenced by surrounding pixels. It is evident from Fig. 7(f) and (g) that the classification results of CMFSL and DM-MRN algorithms exhibit some noise, especially when

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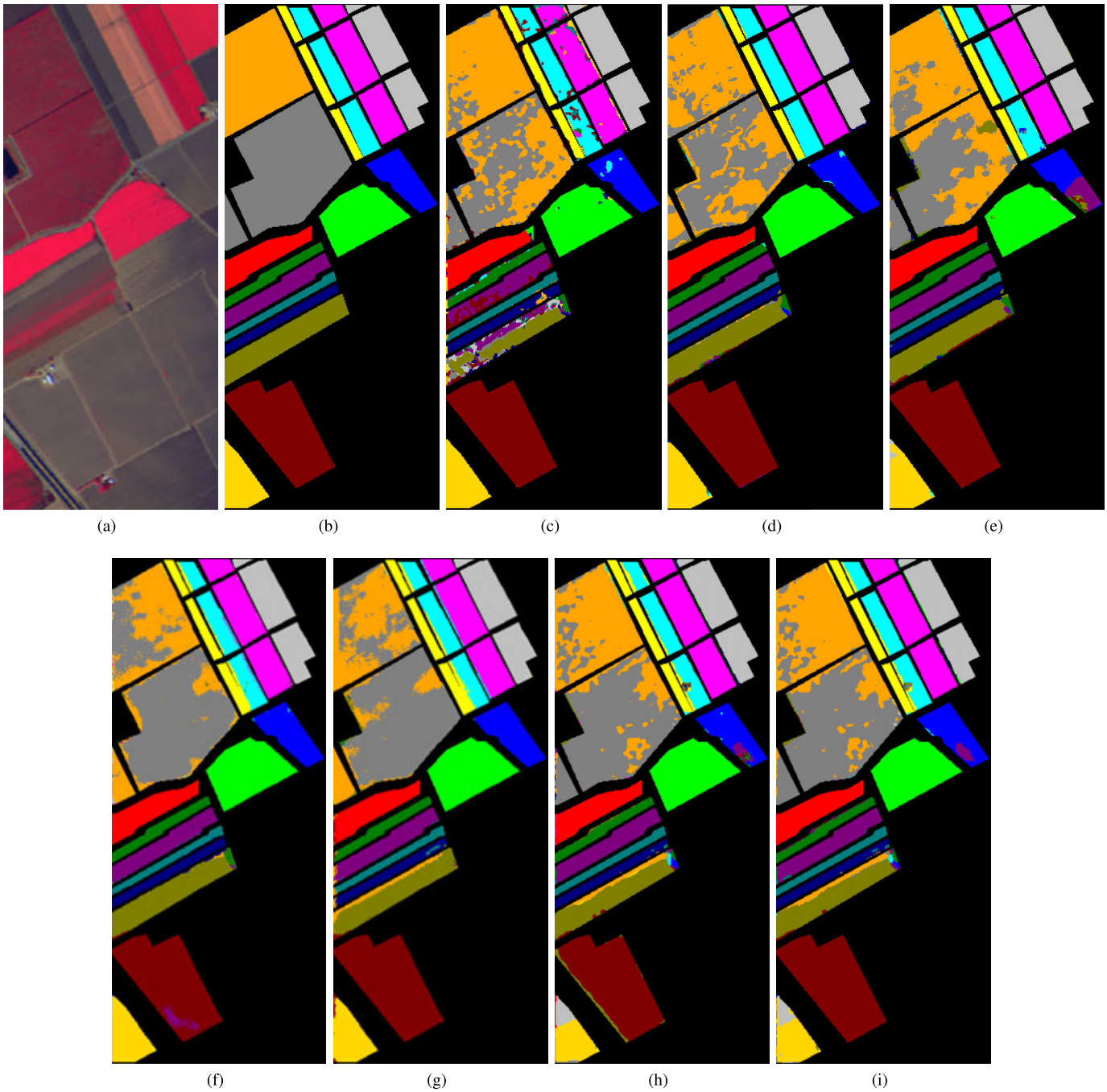


Fig. 8. Visualization maps of various algorithms on the SA dataset. (a) False-color image. (b) Ground truth. (c) SSRN. (d) DFSL+NN. (e) DCFSL. (f) CMFSL. (g) DM-MRN. (h) Gia-FSL. (i) ESAW-FSL.

facing boundaries. The DCFSL and DFSL+NN algorithms in Fig. 7(e) and (d) have more classification errors when facing the 2nd class (Corn note). The result of the Gia-FSL algorithm in Fig. 7(h) has more misclassifications when facing the 11th class (Soybeans min). Compared with the other algorithms, the result of ESAW-FSL in Fig. 7(i) is closer to the ground truth in Fig. 7(b).

The visualization results of 6 algorithms and ESAW-FSL on the SA dataset as the TD dataset are shown in Fig. 8.

As shown in Fig. 8(b), labeled samples are displayed using pixels of different colors, while unlabeled samples are represented as backgrounds using black pixels. It is evident from Fig. 8 that the classification results of most algorithms exhibit some noise in the upper left corner of the image. The classification result map of the SSRN algorithm in Fig. 8(c) has more noise. Based on the class information of the SA dataset provided in Table VII, the boundary classification of SSRN is relatively chaotic when facing the 10th class (Corn senesced green weeds). The result in Fig. 8(e) shows classification errors of the DFSL+NN algorithm when dealing with the eighth class (Grapes untrained). The numerical result of the 15th class is 66.60%, which confirms the dilemma of the DFSL+NN algorithm from another perspective. The classification results of CMFSL and Gia FSL algorithms

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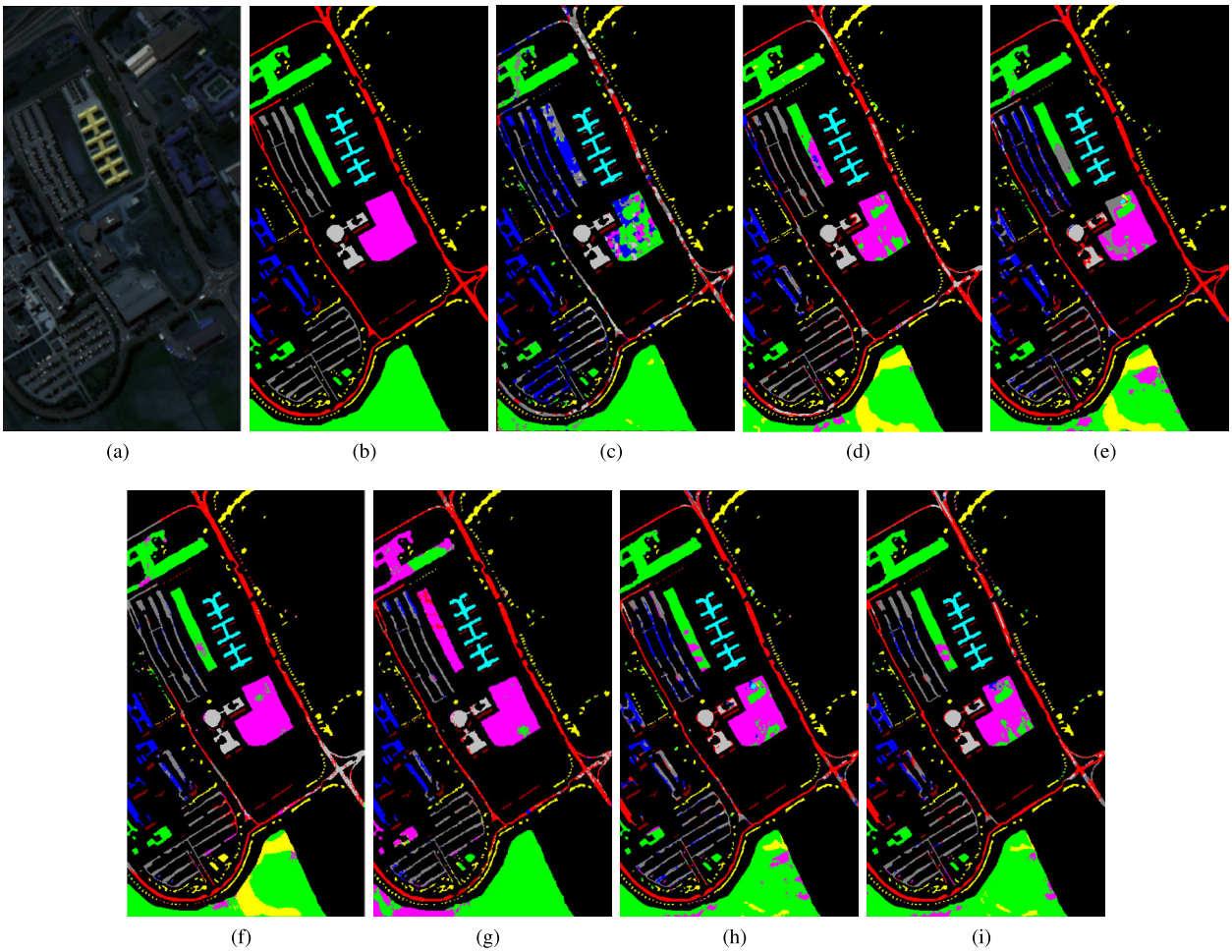


Fig. 9. Visualization maps of various algorithms on the PU dataset. (a) False-color image. (b) Ground truth. (c) SSRN. (d) DFSL+NN. (e) DCFSL. (f) CMFSL. (g) DM-MRN. (h) Gia-FSL. (i) ESAW-FSL.

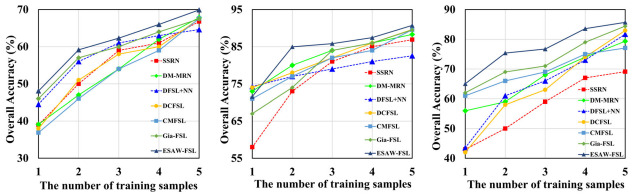


Fig. 10. OA results of three datasets with different training samples. (a) IP. (b) SA. (c) PU.

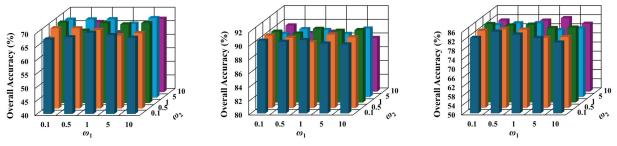


Fig. 11. Experimental results of different ω_1 and ω_2 on three datasets. (a) IP. (b) SA. (c) PU.

based on FSL are similar, and there are errors in boundary classification in the tenth class (Corn sensed green feeds). The classification result of ESAW-FSL is slightly better than those of the compared algorithms.

The visualization results of six state-of-the-art algorithms and ESAW-FSL on the PU dataset as the TD dataset are shown in Fig. 9.

Compared with the ground-truth map in Fig. 9(b), the classification result map of the SSRN algorithm in Fig. 9(c) has noise. Based on the class information provided in Table VI, the SSRN algorithm map is relatively chaotic when facing the eighth class (self-blocking bricks). The DFSL algorithm in Fig. 9(e) shows several classification errors when facing the first class (Asphalt). This may be due to the less dense distribution of the first class samples. The DM-MRN algorithm in Fig. 9(g) classifies part of the samples from the second class (Meadows) as the sixth class (Bare Soil) in the upper left corner. The overall classification result of ESAW-FSL in Fig. 9(i) is better, confirming the effectiveness of ESAW-FSL.

F. Stability Analysis

This article conducts experiments with different training samples on three datasets to verify the stability of ESAW-FSL. The number of training samples is set to [1, 2, 3, 4, 5] in the experiments on the IP, SA, and PU datasets. The stability analysis can be observed through the experimental results between the comparative and proposed algorithms. The experimental results with different training samples are shown in Fig. 10.

From Fig. 10, ESAW-FSL can achieve the best results in most experiments on the three datasets. The DFSL+NN

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algorithm achieves the best OA when the number of training samples is 1 on the SA dataset. This indicates that the ESAW-FSL algorithm is relatively stable when the number changes. The stability is attributed to the embedded self-distillation learning from different perspectives, effective domain adaptation, and DFL for the FSL process. Meanwhile, in Fig. 10, more features can be learned due to the increase in the training samples and the OA results of each algorithm also increase.

G. Parameter Sensitivity Analysis

This article chooses to adjust the weights of the losses in (27) to verify the insensitivity of the proposed algorithm to the loss weights. First, after setting the weight coefficient of DFL to 1 and other parameters, the weight of L_{ESD} ω_1 and the weight of $D_W(T)$ ω_2 are selected from [0.1, 0.5, 1, 5, 10]. The experimental results with different ω_1 and ω_2 are shown in Fig. 11.

In Fig. 11, it can be seen that the range of OA changes is minimal when facing different weights. This is because the experimental results will also change when the parameters change. At the same time, when a certain parameter is fixed, OA fluctuates within a small range and can remain stable within a certain range. Based on the above analysis, OA is not sensitive to variations in ω_1 and ω_2.

IV. Conclusion

This article proposes the ESAW-FSL algorithm to alleviate the problem of limited label samples and the domain shift in cross-domain HSI classification. First, the proposed algorithm utilizes the ESDN to perform embedded self-distillation learning on the SD and TD data, fully extracting practical features from four perspectives. The confidence evaluation of SD features helps extract more instructive information for TD data. To alleviate the problem of domain shift, this article designs a lightweight network to adaptively assist in learning the cost matrix of SD and TD features. This cost matrix can more comprehensively and thoroughly represent domain feature relationships. In the FSL stage, the DFL is used to help hard samples learn more information during the classification process, thereby improving the classification accuracy of hard samples and the network's overall performance. The ESAW-FSL algorithm is compared with six state-of-the-art HSI classification algorithms on the IP, SA, and PU datasets. The results show that the ESAW-FSL algorithm's OA is superior to the comparison algorithms, reaching 69.95%, 90.95%, and 85.70% on the three datasets, respectively. Furthermore, the results of the ablation experiment confirm the effectiveness of each component of ESAW-FSL. Meanwhile, the stability of ESAW-FSL is proved through excellent performance in experiments with different training samples. Of course, there are still areas that need improvement in the proposed method, such as the problem of high computational complexity in implementing domain adaptation in AWD-DA. At the same time, ESAW-FSL processes input data into patch blocks for computation, which reduces the learning of nonlocal information in images. Therefore, efficient utilization of nonlocal information in cross-domain HSI classification is also one of the future research directions.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest in this work.

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